

Module 06: Learning — How Circles Preserve and Pass On Craft

Learning Objectives

1. Explain apprenticeship and legitimate peripheral participation as mechanisms of learning in crochet circles, and connect these to Active Inference concepts of model expansion.
2. Describe how newcomers learn by watching — absorbing the circle's generative model through observation before active participation.
3. Analyze cultural transmission of techniques across generations as a form of generative model propagation.

Introduction

"My grandmother taught my mother, and my mother taught me." This sentence, spoken in crochet circles around the world, captures something profound: crochet is not just a skill — it is a **lineage**. Techniques, patterns, and ways of holding the hook pass from person to person, generation to generation, like a generative model being copied, modified, and extended over time.

In Active Inference, learning is the process of updating a generative model to better predict the world. In a crochet circle, learning happens at multiple scales: the individual learns new stitches, the circle learns what works for the group, and the craft itself evolves as techniques are passed on, adapted, and sometimes lost. This module explores how circles serve as vessels for this multi-scale learning — preserving craft knowledge, transmitting it to newcomers, and evolving it over time.

Key Concepts

1. Legitimate Peripheral Participation

The anthropologist Jean Lave coined the term **legitimate peripheral participation** to describe how newcomers learn in communities of practice. The idea is simple but powerful: you learn by starting at the edges and gradually moving toward the center.

In a crochet circle, this looks like: - **Day one**: The newcomer sits at the table, watches others stitch, listens to conversation, absorbs the pace and culture of the circle. They may not crochet at all, or they may work on simple chains. They are at the **periphery**. - **Week four**: The newcomer is working on a simple project — a dishcloth, a scarf in single crochet. They ask occasional questions. They begin to understand the circle's vocabulary and norms. - **Month three**: The newcomer has moved inward. They contribute to conversations about technique, try new stitches with the group's support, and begin to develop their own style. - **Year one**: The newcomer is now a full participant. They teach newer newcomers, contribute to group decisions, and carry a significant portion of the circle's shared knowledge.

In Active Inference terms, this trajectory is a progressive expansion of the generative model. At the periphery, the newcomer's model of the circle is sparse — they predict little and experience much prediction error. Over time, their model becomes richer, more precise, and more aligned with the group's shared model. The key insight is that this learning does not require formal instruction. It happens through **participation** — through being present, observing, and gradually taking on more complex roles.

The word "legitimate" matters. The newcomer must be genuinely welcome — their participation must be accepted as valid, even when they are only watching or making simple things. Circles that make newcomers feel like outsiders or that demand expert performance from the start undermine this learning process.

2. Learning by Watching

Before a newcomer picks up a hook, they are already learning. **Observational learning** is one of the most powerful mechanisms in a crochet circle:

- **Watching hands:** The newcomer sees how experienced members hold their hooks, tension their yarn, and move through stitch sequences. These observations build a visual and motor generative model — predictions about what the motions should look like — before the newcomer attempts the motions themselves.
- **Watching projects:** Seeing the range of projects at the table (from simple to complex, from conventional to experimental) builds a model of what is possible and what the circle values.
- **Watching interactions:** How people ask for help, how help is given, how mistakes are treated — all of these social observations build the newcomer's model of how to participate in the community.

In Active Inference, this observational learning is a form of **perceptual inference**: the newcomer is building internal predictions by processing sensory evidence from the environment. When they eventually attempt a stitch, they are not starting from zero — they are testing predictions that were formed through hours of watching.

3. Cultural Transmission Across Generations

Crochet circles are one of the primary mechanisms by which craft knowledge crosses generational boundaries. A grandmother's bobble stitch technique, a great-aunt's method for joining squares, a community tradition of making specific items for specific occasions — all of these are **generative models being transmitted across time**.

This transmission is not perfect copying. Each generation adapts what it receives: -

Modification: A daughter learns her mother's granny square but uses different colors, reflecting her own aesthetic generative model. - **Innovation:** A young member learns the traditional join but discovers a new variation that works better for the yarn types available today. - **Loss:** Some techniques are not transmitted — they die with their last practitioner. This is the dissolution of a generative model that no one inherits. - **Revival:** Sometimes a lost

technique is rediscovered through old pattern books or the internet, and re-enters the circle's generative model.

In Active Inference terms, cultural transmission is the propagation of generative models across agents over time. The circle serves as the medium for this propagation — it brings together people of different ages and experience levels, creates the conditions for observation and teaching, and maintains a culture that values the passing on of knowledge.

4. The Circle as a Learning Environment

What makes a crochet circle an effective learning environment? Several features are critical:

- **Safety:** Errors are normalized. Frogging is celebrated as part of the process. This reduces the social cost of prediction error, allowing learners to take risks.
- **Scaffolding:** More experienced members naturally provide support that is calibrated to the learner's level — showing simpler alternatives, offering just enough help without taking over. This is social scaffolding, and it maps directly to the Active Inference concept of adjusting the environment to reduce unnecessary prediction error.
- **Diversity:** As we discussed in Module 2, the range of skill levels in a circle means there is always someone slightly ahead who can model the next step and someone slightly behind whom you can help. This creates a **zone of proximal development** — the sweet spot where learning happens most effectively.
- **Continuity:** The circle meets regularly, providing repeated opportunities for practice, observation, and feedback. Learning in Active Inference requires many cycles of prediction, action, error, and update. The weekly meeting provides this rhythm.

Applications

- **The Grandmother Effect:** In many families, crochet knowledge jumps a generation — a grandmother teaches a grandchild, skipping the parent. This is because the grandchild is at the right stage (beginner) and the grandmother has the right conditions (time, patience, motivation to transmit). The generative model propagates not to the nearest kin but to the kin who is most ready to receive it. This is a reminder that model transmission depends on both sender and receiver.

- **The Internet as Extended Circle:** Online crochet communities — YouTube tutorials, Ravelry forums, Instagram pages — extend the learning mechanisms of the physical circle. A crocheter can observe techniques from thousands of practitioners, ask questions across time zones, and access patterns from traditions they would never encounter locally. In Active Inference terms, the internet massively expands the external states available to the learner, providing an unprecedented variety of prediction errors and model updates. But it lacks the real-time, embodied feedback loop of the physical circle.

Conclusion

Learning in a crochet circle is a multi-layered process: newcomers learn by watching and gradually participating, members teach and learn from each other continuously, and the craft itself is transmitted and evolved across generations. The circle creates ideal conditions for this learning — safety, scaffolding, diversity, and continuity — that enable the progressive expansion of generative models at every scale. This is how craft survives: not in books or patterns alone, but in the living, breathing, stitching communities that carry the knowledge in their hands and pass it on. In the next module, we explore the communicative channels through which this knowledge flows — the stories told over stitches.